BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2018

PHYSICS

Past Paper 2018 • Paper Code: D

Retyped faithfully from the original board paper — same questions, no changes.

SECTION - C (MULTIPLE CHOICE QUESTIONS)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** D

Note: Q. 16: Choose the correct answer from the given options and write the same in your answer copy. Note:- Cutting and overwriting is not allowed.

Q1. Which of these is non –ohm conductors:

(A) Copper
(B) Silver
(C) Iron
(D) Semi conductor

Q2. To measure the current in a circuit an ammeter is always connected in:

(A) Series
(B) Parallel
(C) Neither parallel nor series
(D) Some time parallel some time series

Q3. Which one of the following is called inertia of electricity

(A) Back - emf
(B) Solar cell
(C) Potential gradient
(D) None of these

Q4. The resistors of resistance 2 Ohms, 5 Ohms and 7 Ohms are connected in parallel. If the potential difference across 5 Ohms is 5 volt, the potential difference across other resistors is:

(A) 9 volt
(B) 5 volt
(C) 2.5 volt
(D) 1.5 volt

Q5. Television announced 48° temperature of Sukkur in cold January day, this temperature corresponds to:

(A) Fahrenheit
(B) Celsius
(C) Kelvin
(D) All of these

Q6. The mass of a particle moving with velocity of light is equal to: What?

(A) Zero
(B) Infinite
(C) Finite
(D) Unit

Q7. The velocities of seven molecules are 1 Km/h, 2Km/h, 3Km/h, 4Km/h, 5 Km/h, 6Km/h and 7 Km/h respectively. Find the mean square speed of these gas molecules.

(A) 20 Km/h
(B) 100 Km/h
(C) 40 Km/h
(D) 140 Km/h

Q8. Work function for a metal is 4.14 ev. Threshold frequency of the incident light for the metal is:

(A) 10^{12} Hz
(B) 10^{15} Hz
(C) 10^{13} Hz
(D) 10^{16} Hz

Q9. Three dimensional images of objects obtained by using lasers in a process called:

(A) Crystalography
(B) Holography
(C) Geography
(D) None of these

Q10. In a p – n junction diode the width of depletion layer is 0.5 μm and potential barrier is 0.5 volt. Calculate the barrier electric field:

(A) 10^{-6} V/m
(B) 10^{4} V/m
(C) 10^{6} V/m
(D) None of these

Q11. An electron passing close to a target nucleus there by slows down by radiating some of its energy, this process is called:

(A) Compton effect
(B) Photo electric effect
(C) Bremsstrahlung
(D) Laser

Q12. Which one of the following quantities in electricity is analogous to temperature?

(A) Potential
(B) Resistance
(C) Charge
(D) Heat

Q13. A proton falls freely under gravity. What will be de – Broglies wavelength after 10 seconds of its motion. (Take g=10 ms^{-1})

(A) 3×10^{-8} m
(B) 6.63×10^{-34} m
(C) 19.6×10^{-14} m
(D) 3.95×10^{-9} m

Q14. Which term relates to magnetism:

(A) Ohm
(B) mole
(C) Gauss
(D) Temperature

Q15. When the p-n junction is forward – biased the width of depletion region.

(A) Increases
(B) Decreases
(C) Remains same
(D) p-n junction cannot be forward-biased

Q16. An electric eye operates because of:

(A) Compton effect
(B) ɣ – radiation
(C) Radiation pressure
(D) Photoelectric effect

Q17. The quantity that remains unchanged in a transformer is:

(A) Voltage
(B) Current
(C) Potential difference
(D) Frequency

SECTION – A (Short Answer Questions)

Time Allowed: 2 Hours and 40 minutes | **Maximum Marks:** 40

Note: NOTE: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. What is meant by the term “Mutual Induction”? Deduce an expression for mutual inductance between a pair of coils.

Q2. Which law is obeyed by a gas during isothermal process? State and express that law:

Q3. How coulomb's law can be expressed in mathematical form by introducing a dielectric material between two point charges?

Q4. Calculate the energy of a photon whose frequency is 20 MHz.

Q5. Briefly explain capacitor. Why the capacitance of a parallel plate capacitor increases on introducing a dielectric medium between the plates?

Q6. What is equivalent resistance of three 5 Ω resistors connected? (i) in series (ii) in parallel.

Q7. How can a galvanometer be converted to voltmeter to read maximum potential difference. Support your answer with related mathematical expression.

Q8. A single Diode rectifier provides half wave rectification. Express with required labelled diagram.

Q9. Prove that $\lambda = h / \sqrt{2Vem}$

Q10. State mass defect and binding energy by showing graphical interpretation of nuclear stability.

Q11. What are harmful effects of radiations? What measures can be adopted to safeguard from the nuclear hazards?

SECTION – B (LONG ANSWER QUESTIONS)

Maximum Marks: 28

> Note: NOTE: ATTEMPT ANY TWO (02) QUESTIONS.

Q12.

(a) [09] State and explain Ampere's law. Show how the equation for Ampere's circuital law is modified in the presence of current in toroid. Illustrate with diagram.

(b) [05] A solenoid 1 m long has two layers of windings 1000 turns each and carries a current of 5 amperes. Find magnetic induction at its centre along axis.

Q13.

(a) [09] What do you know about Black Body Radiation? Draw neat and labelled diagram of energy distribution graph in terms of wavelength. Express the required laws of it.

(b) [05] What will be de-Broglie wave length of a mass 6 Kg moving with velocity of 1000m/s

Q14.

(a) [09] State and prove Gauss law. Obtain an expression for electric field intensity at a point outside a charged conducting sphere and field at a point on the surface.

(b) [05] A particle carrying a charge of 10^{-5} c starts from rest in a uniform electric field of intensity 50 Vm^{-1}. Find the force on the particle and the kinetic energy it acquires when it has moved 1m.